

[Collaborative Docs](#) >

Discussion Agenda 08/14/2016

Below is a discussion agenda / live progress report for the Collaboration-wide brainstorming session of the 5th AGORA Workshop @ University of California Santa Cruz (Aug. 12–14, 2016; see the [program](#)).

During this year's workshop, we focus on finalizing the paper being pushed by Paper Group 4 – Isolated Disk ([workspace](#)).

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Check out the important items in red color below!

(1st revision: 08/15/2016)

[1. Progress Report \(as of Aug. 2016\)](#)

We are actively making progress, chiefly, in three different areas below (numbered tentatively).

	PAPER "2"	PAPER "3"	PAPER "4"																																																				
GOALS	• Run Proof-of-concept-DM (~1e11 Ms halo) at higher resolution	• Implement and test common astrophysics package • Run Proof-of-concept-NOSUBGRID (~1e11 Ms halo)	• Compare simulation codes using common disk ICs • Test common astrophysics models including Grackle																																																				
LEADERS	• Oliver Hahn (OCA, Nice) • Christine Simpson (HITS)	• Nick Gnedin (Fermilab) • Piero Madau (UCSC) • Britton Smith (Edinburgh)	• Oscar Agertz (Surrey) • Romain Teyssier (Zurich) • James Wadsley (McMaster)																																																				
PARTICIPANTS	• Working Groups engaged: - III (common cosmo ICs) - IV (common analysis) - VI (dwarf galaxies) • Code representatives:	• Working Groups engaged: - I (common physics) - III (common cosmo ICs) • Code representatives:	• Working Groups engaged: - II (common disk ICs) - IV (common analysis) - V (subgrid physics) • Code representatives:																																																				
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PROGRESS MADE OR BEING MADE	• Please visit Paper 2's Workspace where Christine Simpson, Oliver Hahn and others have been updating the progress. • We will plan to have 5-code comparison. As of Aug. 2016, 3+ participating groups have already completed and submitted the production-level results (~100 pc resolution) to NERSC.	• Either via Britton's original interface or through Nick's API, GRACKLE is now fully implemented (equilibrium cooling version) in all 9 participating codes. • Code groups such as ART-I and GIZMO have carried out cosmological simulations with various AGORA ICs, even including sophisticated subgrid physics. • Let us consider running at least some codes with Grackle's non-equilibrium cooling mode.	• Please visit Paper 4's Workspace where Oscar Agertz, Romain Tessier and others have been updating the progress. • As of Aug. 2016, all 9 participating code groups have submitted their "first" (Grackle + noSF) and "second" datasets (Grackle + SF + ThermalFbck). Further revision to an already submitted dataset may be discouraged at this point. • The latest composite figures on their Workspace were created on Jul. 20, 2016. A draft of Paper 4 is being finished during the 5th AGORA Workshop. • Ideas for Paper "4b" are being formulated; please see Section 2 below. • The agora-analysis-script repository is being accordingly updated as the analysis progresses.																		
OTHER LINKS	• Please see the progress report in Paper 2's Workspace and/or in WG III's Workspace. • MUSIC Google group	• GRACKLE Google group • GRACKLE download & documentation • AGORA API for GRACKLE by Nick Gnedin	• Please see the progress report in Paper 4's Workspace and/or in WG V's Workspace. • yt-AGORA Google group																		

[2. Paper Group 4 - Progress and Discussion](#)

Below is an extremely shortened summary of discussion from [this year's Workshop](#) only for this telecon. The writings below will not be updated as the ideas in Paper "4" evolve in time. Participants of Paper "4" are STRONGLY encouraged to check out and follow the up-to-date consensus in the [Paper 4 Workspace](#).

(1) Timeline To Submit Paper "4" and Afterwards

- 2+ code groups will likely have to rerun the simulation in the next week.
- **Circulate a finalized form of the draft within a month.**
- Submit the paper in September, in time for various upcoming funding proposals this fall.
- **Next paper (a.k.a. **Paper "4b"**) using the existing isolated runs and analysis pipeline**
 - To be led by (tentatively) Santi Roca-Fabrega
 - Analysis items that were not covered in Paper "4" : see [this link](#)
 - 500 Myr snapshots to be publicly released in Jan. 1, 2017 (Joel's suggested date; speak up if you disagree)

(2) Discussed : Paper "4" Authorship

- Order of author list: Multi-tiered approach, 3 first authors (Kim, Teyssier, Agertz) + code leaders + Grackle developers + the rest, Ordered alphabetically in the 2nd, 3rd, and 4th tiers
- Institutions can go with numbers while details about code leaders or Grackle developers with letters.
- Final author list and its order to be confirmed by the AGORA Steering Committee

(3) Finishing Up Paper "4"

- Section 6 is being written during the weekend working session. Please visit the [collected links](#) for this year's Workshop.
- **Learn what was discussed in the Workshop by finding the Google Doc in the link above. Also in the same link, learn how you could participate in the writing of "Section 6. Results". The Section 6's text will be open to your contribution until Aug. 21.**
- The Google Doc above also transparently describes all other topics discussed during the Workshop.
- Note that the draft you will find in the link above (in a pdf form; last updated Aug. 12) will significantly change after Aug. 21 with reorganized and revamped sections, new figures, new styles, etc.

[3. Other Topics for Collaboration-wide Discussion](#)

(1) Updates from Paper Group "2" – Higher-resolution DM-only Cosmological Run

- See their [Workspace](#) for more information.
- Tentative deadline to submit the datasets : Sep. 30, 2016
- Joel will check with Anatoly about a possible ART-I involvement, and with Peter about a integration of Rockstar and consistent tree in yt.
- Consistent tree – python interface by Britton Smith : visit [this link](#)

(2) Next Papers to Focus

- Paper "2", then Paper "3" and Paper "4b"

(3) Future Timeline of the Project in General

- Multiple funding opportunities to be sought after this fall.

[4. Recaps](#)

(1) AGORA SMBH Prescription (Accretion and Feedback):

- Please visit [this link](#).

(2) Strategy to Store Simulation Outputs:

- Please visit [this link](#).

(3) Your Own yt Installation on NERSC:

- Please visit [this link](#).

(4) yt–Sunrise Pipeline for AGORA on NERSC:

- Please visit [this link](#).

(5) Mini–workshop Opportunities:

- Mini–workshop for individual paper groups to write papers: KIPAC@Stanford is willing to provide funding opportunities for working group leaders to organize mini–workshops (~10 people). An example was the [yt–AGORA mini–workshop](#) at UCSC in Mar. 2014.

(6) Useful Links:

- Quick links to the important pages in AGORA Workspace, including internal policies: [here](#)
- Internal policy regarding the disclosure of AGORA simulation results: [here](#)
- Collection of softwares relevant to AGORA effort: [here](#)

[5. Your Feedback](#)

Let us hear anything AGORA, including the scientific and organizational effort currently being made. Voice your opinion during the web conference! Feel free to share your thoughts in the comments section below, or on the [mailing list](#). You can also directly contact the [project coordinator](#) anytime!